

1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

CS XI

Chapter 1: Computer Systems

Lecture 06 of 12

★ Golden

1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams

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LEARN IT

A **logic expression** (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$. To evaluate it correctly you must follow the **order of precedence**, just like BODMAS in maths:

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

WORKED EXAMPLE

Worked example 1: truth table of $Y = A + B' \cdot C$

Step 1: 3 inputs, so 8 rows.

Step 2: Precedence: first find B' (NOT), then $B' \cdot C$ (AND), then $A + (B' \cdot C)$ (OR). Make a column for each step.

A	B	C	B'	$B' \cdot C$	$Y = A + B' \cdot C$
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Step 3: $Y = 1$ in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand: $A=0, B=0, C=1 \rightarrow B' = 1, B' \cdot C = 1, Y = 0 + 1 = 1$. ✓

WORKED EXAMPLE

Worked example 2: truth table of $Y = (A + B) \cdot C'$

Step 1: Parentheses first: find $A + B$. Then NOT: C' . Then AND the two.

A	B	C	A+B	C'	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Step 2: $Y = 1$ in rows 2, 4 and 6 (decimal).

Minterms (SOP) and Maxterms (POS)

LEARN IT

Sometimes you are given a truth table and must *find* the expression. There are two standard ways.

Minterm (used in SOP — Sum of Products): a **product (AND)** of all the variables, written for a row whose output is **1**. A variable is **complemented (A')** if its value is 0, and **uncomplemented (A)** if its value is 1. OR-ing all the minterms gives the SOP expression.

Maxterm (used in POS — Product of Sums): a **sum (OR)** of all the variables, written for a row whose output is **0**. A variable is **complemented (A')** if its value is 1, and **uncomplemented (A)** if its value is 0. AND-ing all the maxterms gives the POS expression.

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	–	$A + B$
0	1	1	$A'B$	–
1	0	0	–	$A' + B$
1	1	1	AB	–

Final SOP expression: $Y = A'B + AB$. **Final POS expression:** $Y = (A + B)(A' + B)$. Both describe the same table. (Both simplify to $Y = B$ — look at the table: Y always equals B. You will confirm this with a K-map later.)

WORKED EXAMPLE

Worked example 3: write the SOP for a 3-input "majority" circuit. Y = 1 when at least two of A, B, C are 1.

Row	A	B	C	Y	Minterm
0	0	0	0	0	–
1	0	0	1	0	–
2	0	1	0	0	–
3	0	1	1	1	A'BC
4	1	0	0	0	–
5	1	0	1	1	AB'C
6	1	1	0	1	ABC'
7	1	1	1	1	ABC

Step 1: Pick the rows with Y = 1: rows 3, 5, 6, 7.

Step 2: Row 3 is A=0, B=1, C=1 → A is 0 so write A'; B and C are 1 so write B, C → A'BC.

Step 3: Do the same for the other rows and OR them:

$Y = A'BC + AB'C + ABC' + ABC$. In short form: $Y = \Sigma m(3, 5, 6, 7)$. We will simplify this with a K-map in 1.1.13.

$$Y = A'BC + AB'C + ABC' + ABC = \Sigma m(3, 5, 6, 7)$$

REMEMBER IT

SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

COMMON MISTAKE

Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where Y = 1; (2) forgetting precedence — in $A + B \cdot C$ you must do $B \cdot C$ first. $A + B \cdot C$ is **not** $(A + B) \cdot C$: for A=1, B=0, C=0 the first gives 1 and the second gives 0.

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR۔ منٹرم (Minterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 1 ہو؛ اس میں 0 والے متغیر پر بار (A) لگتی ہے۔ میکس ٹرم (Maxterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 0 ہو؛ اس میں 1 والے متغیر پر بار لگتی ہے۔

CHECK YOURSELF

1. Which comes first: AND or OR?
2. Write the minterm for $A = 1, B = 0, C = 1$.
3. Write the maxterm for $A = 0, B = 1$.

Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.

LEARN IT

A **logic diagram** shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression. To draw one, use the order of precedence: the operation with the **highest priority is drawn first** (nearest the inputs), and the last operation is the gate that gives the final output.

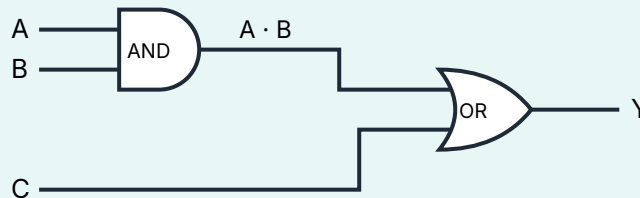
WORKED EXAMPLE

Example 1: Draw the logic diagram for $Y = A \cdot B + C$.

Step 1: Find the operations: AND ($\cdot$) and OR ($+$).

Step 2: AND has higher precedence than OR, so draw the AND gate first, with inputs A and B. Its output is $A \cdot B$.

Step 3: Draw an OR gate. Feed it $A \cdot B$ and C. Its output is Y.



$Y = A \cdot B + C$: AND first, then OR.

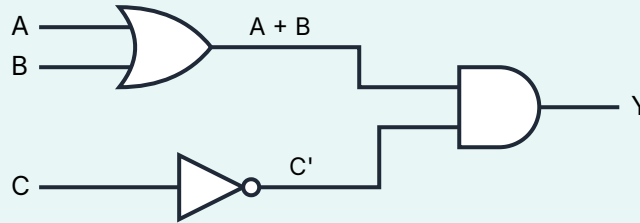
WORKED EXAMPLE

Example 2: Draw the logic diagram for $Y = (A + B) \cdot C'$.

Step 1: Parentheses have the highest priority, so draw the OR gate for $A + B$ first.

Step 2: NOT is next: draw a NOT gate on input C to get C' .

Step 3: The last operation is AND: feed $(A + B)$ and C' into an AND gate to get Y .



$Y = (A + B) \cdot C'$: OR and NOT first, then AND.

You can check this circuit against the truth table in 1.1.8 (worked example 2): $Y = 1$ only for rows 010, 100 and 110.

REMEMBER IT

Read the expression and ask "What is done **last**?" That gate goes on the far right, next to Y . Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2 gates; $Y = (A+B) \cdot C'$ needs 3.

COMMON MISTAKE

Do not draw a separate wire for a variable each time it appears and then leave one unconnected. Every gate input must be joined to an input variable or to another gate's output, and every wire should be clearly labelled in exams.

اردو میں سمجھیں

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔ ترجیحی ترتیب کے مطابق جو آپریشن پہلے ہوتا ہے اس کا گیٹ پہلے (ان پٹ کے قریب) بنایا جاتا ہے، اور آخری آپریشن کا گیٹ آؤٹ پٹ Y دیتا ہے۔ مثلاً $Y = A \cdot B + C$ میں پہلے AND گیٹ اور پھر OR گیٹ بنتا ہے۔

EXAM TIP

Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols with labelled intermediate outputs (like $A \cdot B$), and finish with the truth table. Marks are lost for wrong gate shapes and missing bubbles.

CHECK YOURSELF

1. In $Y = A + B \cdot C$, which gate is drawn first?
2. How many gates are needed for $Y = A' \cdot B$?

Answers: 1. The AND gate (for $B \cdot C$). 2. Two: one NOT gate (for A') and one AND gate.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Precedence (highest first) is:

- a. () then NOT then AND then OR
- b. OR then AND then NOT
- c. AND then () then OR
- d. XOR first always

MCQ 2. A minterm is used when Y is:

- a. 1 (SOP)
- b. 0 (POS)
- c. Always 0
- d. A stack

MCQ 3. For $AB=01$ if $Y=1$ the minterm is:

- a. $A'B$
- b. AB
- c. $A+B$
- d. $A'B'$

MCQ 4. A maxterm is a:

- a. Sum (OR) for a 0 output
- b. Product for a 1
- c. K-map only
- d. Gate delay

MCQ 5. SOP means:

- a. Sum of products
- b. Sum of people
- c. Stack of pointers
- d. Set of pies

MCQ 6. In a logic diagram you draw first the operation with:

- a. Highest precedence
- b. Lowest precedence
- c. Most inputs
- d. Green ink

Extra short questions

1. State operator precedence in Boolean algebra.
2. Define minterm and maxterm.
3. From Y rows 01 and 11 = 1, write SOP.
4. For $Y = A \cdot B + C$, which gate is drawn first?

Extra long question

1. Given a 2-input truth table with $Y = 0,1,0,1$, write minterms, SOP, maxterms that apply, and draw the logic diagram.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Parentheses, NOT, AND, OR.
2. Minterm: AND of all variables (complemented if 0) for rows where $Y=1$. Maxterm: OR of all variables (complemented if 1) for rows where $Y=0$.
3. $Y = A'B + AB$.
4. AND ($A \cdot B$), then OR with C, because AND beats OR.

Long

1. $Y=1$ on 01 and 11 $\rightarrow A'B + AB$ which simplifies to B. Diagram: if unsimplified, two ANDs into an OR; simplified, $Y=B$ (wire). Show both and say examiners accept the simplified form if working is shown.

Next lecture: 1.1.13 Karnaugh Maps (K-Maps) Simplification